

ELKO, NV, USA

WMO: 725825

Lat: 40.829N

Lon: 115.789W

Elev: 5030

StdP: 12.21

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24121

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.1	(c) 4.5	(d) -6.5	(e) 4.7	(f) 2.5	(g) -1.6	(h) 6.1	(i) 11.0	(j) 23.8	(k) 39.7	(l) 19.5	(m) 38.7	(n) 3.3	(o) 80	(p) 0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.8	95.6	59.1	93.0	58.3	90.4	57.4	62.0	86.7	60.6	85.1	59.4	83.8	8.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.5	76.1	66.8	51.5	68.1	67.3	48.7	61.4	67.1	30.0	86.4	29.0	84.3	28.0	83.4	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	18.7	16.3	-8.6	99.3	6.5	3.0	-13.3	101.5	-17.1	103.2	-20.8	104.9	-25.5	107.0
			-9.2	64.8	6.4	2.1	-13.8	66.2	-17.5	67.5	-21.1	68.6	-25.7	70.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	26.7	30.6	39.4	45.1	53.6	63.1	72.0	69.4	59.8	46.9	35.9	26.9		
	DBStd	17.25	9.75	8.76	7.24	7.07	7.47	7.28	5.16	4.93	7.34	7.72	8.72	9.13		
	HDD50	3134	721	543	335	177	48	6	0	0	14	149	425	716		
	HDD65	6835	1186	963	795	596	357	115	8	18	181	562	873	1181		
	CDD50	2240	0	0	5	31	161	399	681	600	308	53	2	0		
	CDD65	467	0	0	0	0	5	58	224	153	26	1	0	0		
	CDH74	10943	0	0	1	27	326	1778	4265	3348	1112	86	0	0		
	CDH80	5239	0	0	0	3	73	753	2322	1693	383	12	0	0		
(14) Wind	WSAvg	5.5	4.8	5.5	6.1	6.8	6.4	6.3	5.8	5.4	5.0	4.8	4.7	4.7		
(15) Precipitation	PrecAvg	9.9	1.1	0.8	0.9	0.9	1.1	0.8	0.4	0.6	0.6	0.7	1.1	1.1		
	PrecMax	18.3	3.3	2.3	2.4	2.2	4.1	2.6	1.5	4.6	3.2	2.0	2.8	4.2		
	PrecMin	4.8	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0		
	PrecStd	3.0	0.8	0.5	0.6	0.6	0.9	0.7	0.4	0.8	0.7	0.6	0.6	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.4	60.3	70.1	77.6	86.4	95.8	99.7	96.4	90.9	81.6	67.2	53.3		
		MCWB	40.7	43.2	46.9	51.2	55.8	59.3	61.5	59.1	57.4	53.4	47.3	41.4		
	2%	DB	47.9	53.7	65.0	72.5	81.5	91.2	97.0	93.9	87.6	75.8	61.8	47.9		
		MCWB	38.5	41.3	45.2	48.8	54.3	56.9	59.9	58.1	55.5	50.3	44.8	39.2		
	5%	DB	44.2	48.5	60.6	67.7	77.6	88.0	94.5	91.7	84.6	71.5	56.4	44.4		
		MCWB	36.8	38.7	43.6	46.9	52.2	56.1	59.0	57.5	54.5	48.2	42.3	36.9		
	10%	DB	40.1	44.1	55.7	62.8	72.9	84.4	91.6	89.6	81.3	66.5	51.6	40.5		
		MCWB	34.6	36.5	41.6	45.2	50.7	55.1	58.2	56.9	53.8	46.5	40.4	34.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.3	44.8	48.5	52.9	58.2	61.3	64.4	62.8	61.1	54.6	49.0	43.4		
		MCDB	48.9	56.8	66.7	74.8	81.7	89.9	91.5	85.5	80.1	77.5	62.6	50.2		
	2%	WB	39.8	41.9	46.1	49.8	55.3	59.1	62.6	61.1	58.5	52.0	45.6	40.5		
		MCDB	46.0	51.9	62.8	68.9	76.7	85.9	88.1	84.6	78.9	71.2	59.2	46.7		
	5%	WB	37.2	39.4	44.1	47.8	53.5	57.4	61.4	59.8	56.7	49.7	43.3	37.5		
		MCDB	42.7	47.1	59.2	65.2	73.3	83.7	86.5	83.8	79.1	67.0	55.2	43.1		
	10%	WB	35.0	37.1	42.0	45.8	51.7	56.1	60.1	58.5	54.7	47.6	41.1	35.2		
		MCDB	39.8	43.8	54.6	61.3	70.4	81.3	85.0	83.2	77.9	64.6	50.2	39.8		
(35) Mean Daily Temperature Range	MDBR		20.9	21.5	25.7	26.5	29.5	34.7	37.8	38.6	37.3	33.9	26.7	20.9		
	5% DB	MCDBR	25.2	26.4	35.0	38.0	38.5	41.4	40.7	41.6	42.4	42.6	34.8	24.1		
		MCWBR	17.1	16.0	18.2	18.1	16.1	15.9	14.9	16.4	18.7	22.0	20.3	16.4		
	5% WB	MCDBR	21.7	24.2	32.2	33.7	34.2	38.3	35.2	36.0	36.6	37.3	32.0	20.1		
		MCWBR	15.1	15.2	17.3	16.4	14.8	15.2	13.4	14.5	16.2	19.4	19.0	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.244	0.258	0.271	0.299	0.309	0.311	0.330	0.320	0.300	0.271	0.255	0.245		
	taud		2.528	2.552	2.562	2.476	2.495	2.518	2.482	2.486	2.542	2.614	2.600	2.555		
	Ebn at Noon		299	310	315	309	306	305	298	298	300	301	293	291		
	Edh at Noon		22	25	29	33	34	33	34	33	29	24	21	20		
(44) All-Sky Solar Radiation	RadAvg		640	927	1326	1728	2079	2473	2384	2133	1756	1220	772	524		
	RadStd		85	118	101	120	213	168	133	123	91	83	57	90		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.08	N/A	N/A	+1.91	N/A	N/A	N/A	-279	+365	+196
(47)	Regional (5 neighbors)	N/A	N/A	N/A	+1.10	N/A	N/A	N/A	N/A	+218	+103

Nomenclature: See separate page